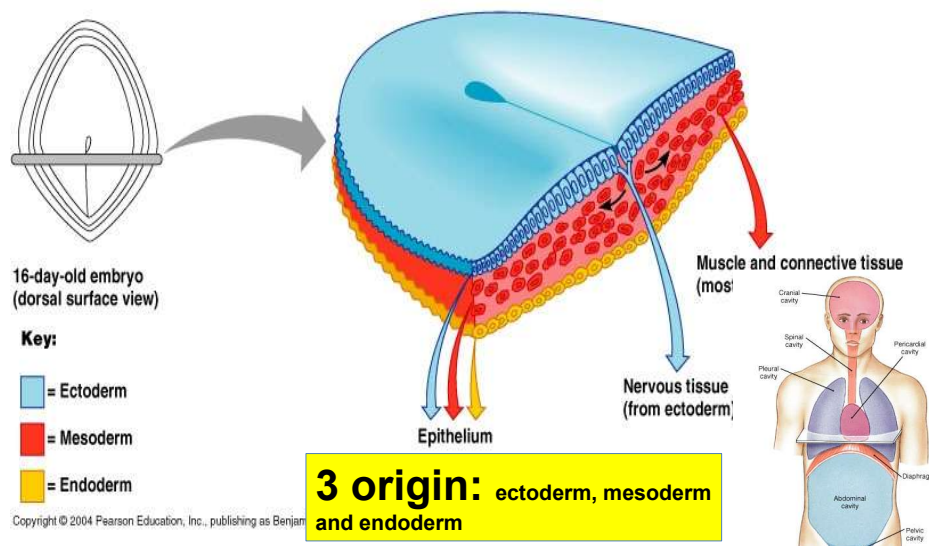


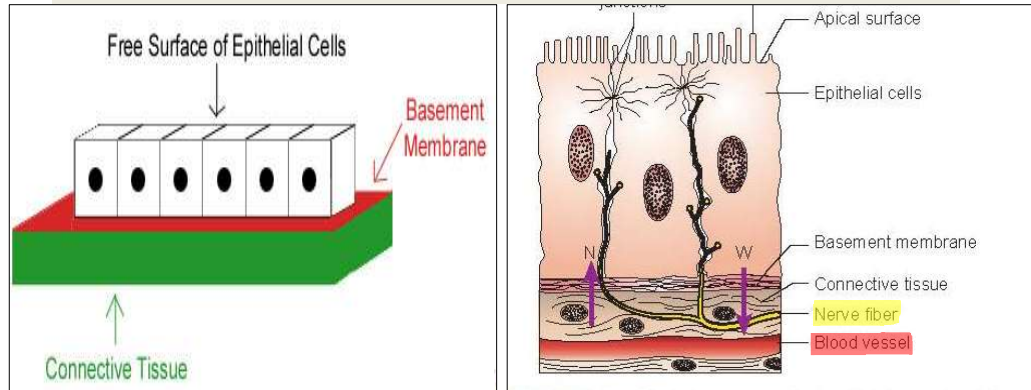
Epithelial Tissue

by
PROF. Eman Abas

General Characters of Epithelial Tissue

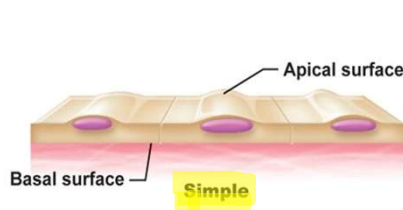


- Crowded cells.
- Resting on a basement membrane.
- High power of regeneration.
- **A vascular,**
- Nutrition by diffusion from **CT**

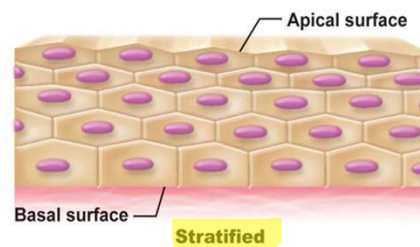


SURFACE EPITHELIUM=....

Classified according To **Number of Layers :**



=1 layer of cells
resting on basement
membrane [BM]

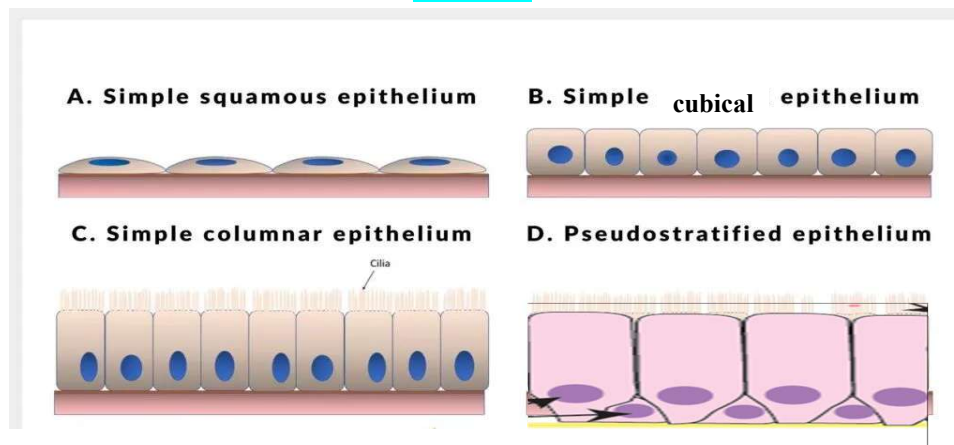


= More than 1 layer
of cells

SIMPLE EPITHELIUM=....

1 layer of cells resting on basement membrane [BM]

Types : shape of cells??



Simple Squamous:

Str: One layer of flat cells with flat nuclei on BM.

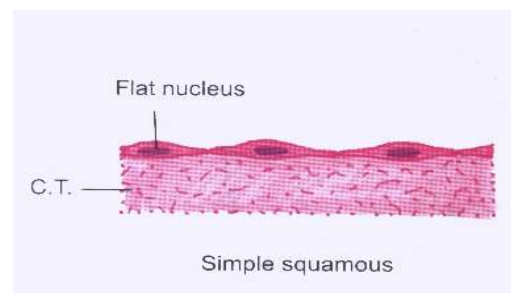
thin

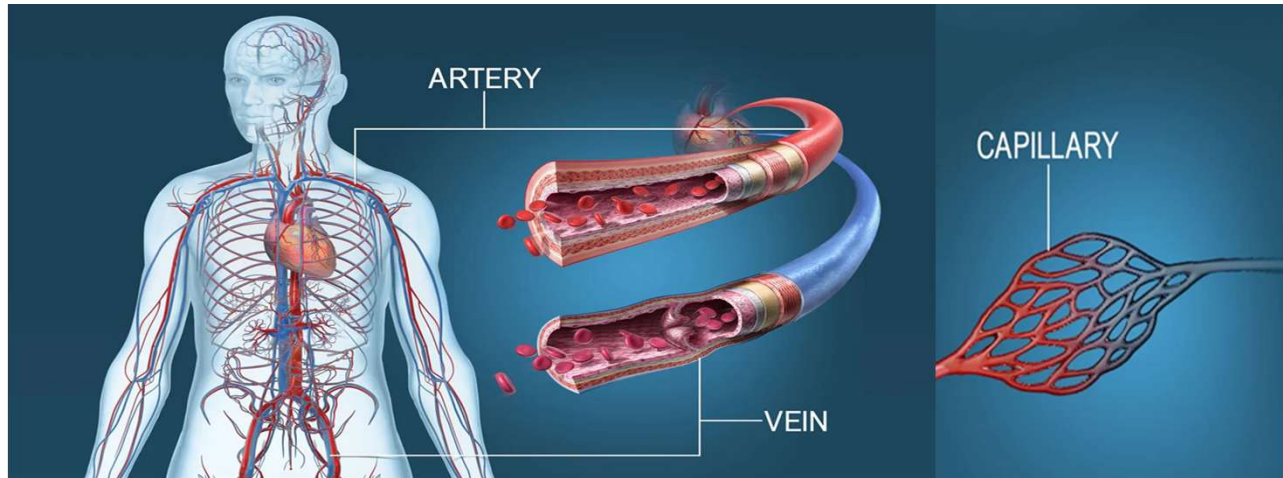
smooth

Sites&functions:

1. Alveoli of lung = ? **thin**
2. Bowman's capsule ? **Thin**
3. Heart & bl. v = ? **Smooth**
4. Serous membranes (pleura pericardium-peritoneum) = ? **smooth**

A. Simple squamous epithelium





Vascular system

By Prof. Eman Abas

Blood Vessels types

Arteries:

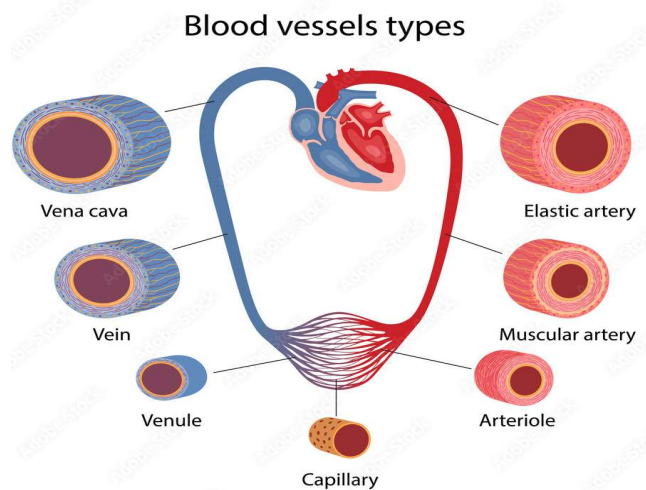
Large, medium & arterioles.

Veins:

Large , medium & venules.

Arterio-venous connections:

-Blood capillaries

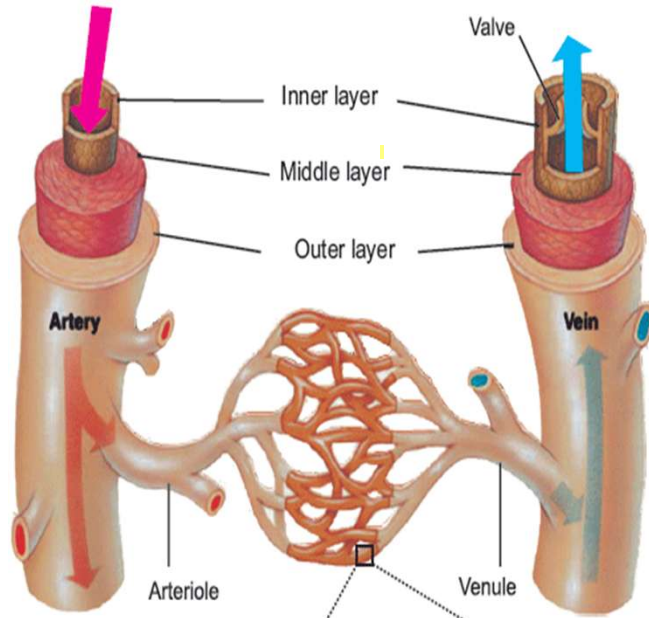


General structure of blood vessels

1-Tunica Intima: Innermost layer, contact with blood.

2-Tunica Media: middle layer of blood vessels.

3-Tunica Adventitia: outermost layer



1- Tunica Intima: innermost

Fn: provides a smooth surface for bl. flow
formed of:

a- Endothelium:

=Simple squamous. epith?

smooth

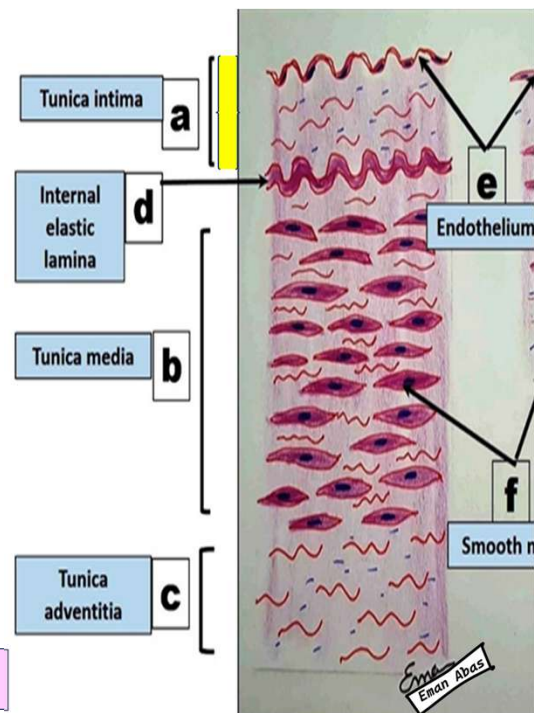
b- Subendothelium CT:

Loose areolar C.T.? support

c-Internal elastic lamina:

only in arteries

=layers of elastic fibers? ≠ occlusion



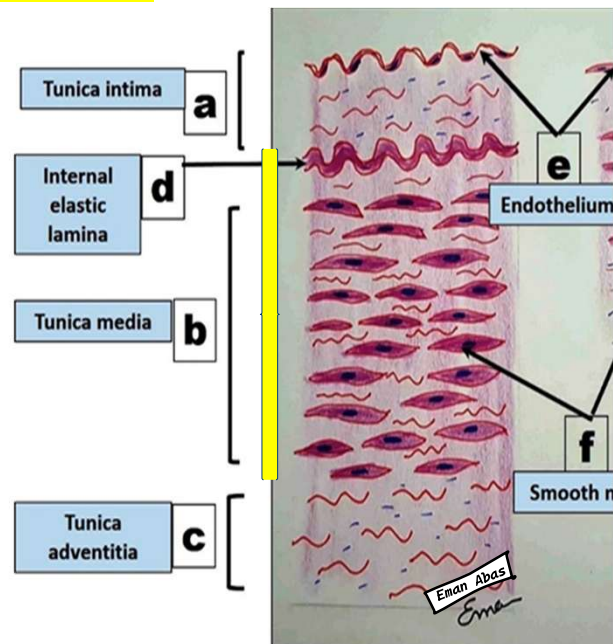
2- Tunica Media: middle layer

fn: Regulates blood flow

formed of:

1. **SMF** **Circularly** arranged
=Smooth muscle fibre
→ **Control diameter**
2. **Elastic fibers.**
→ **Allow distension**
3. **± External elastic lamina**

in arteries

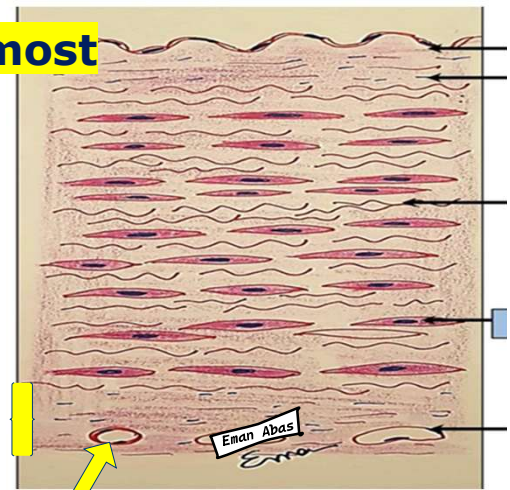


3- Tunica Adventitia: outermost

- **Formed of** loose C.T. with Collagen fibres (**longitudinal**) + elastic fibres+ C.T. cells.

Contains nerves, lymphatics and vasa vasorum

- **FN:**
- **Connect** BV to the surrounding tissues
- **Prevent** over-distension of BV.



▪ **Vasa Vasorum:**
blood vessel of the blood vessel
– Small arteries mainly in Adventitia of large vessels to nourish OUTER PART

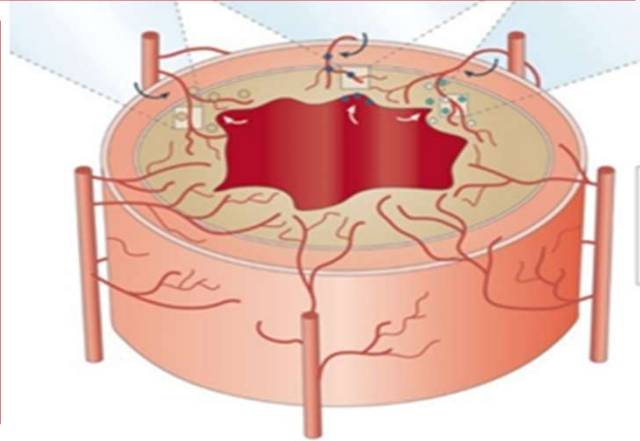
Why vasa vasorum are present in the tunica adventitia of large thick-walled arteries and in veins?

❖ *large arteries?????*

to nourish the outer part of the thick vessel wall

❖ *veins ????*

blood is deoxygenated.



Arteries

- Classified according to
- size & main tissue in media into:

1- Large elastic arteries:

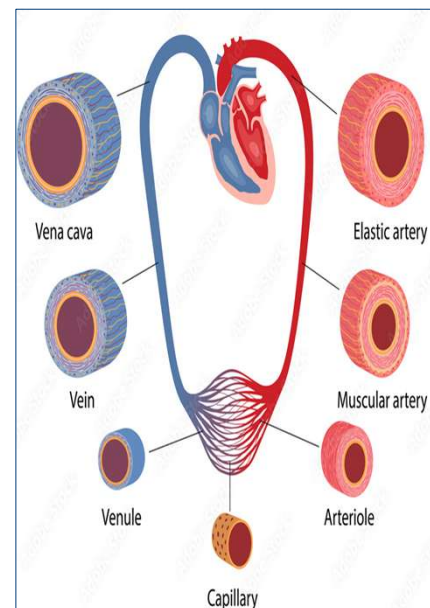
carry blood *from heart*.

2- Medium-sized muscular arteries:

carry blood *to organs*.

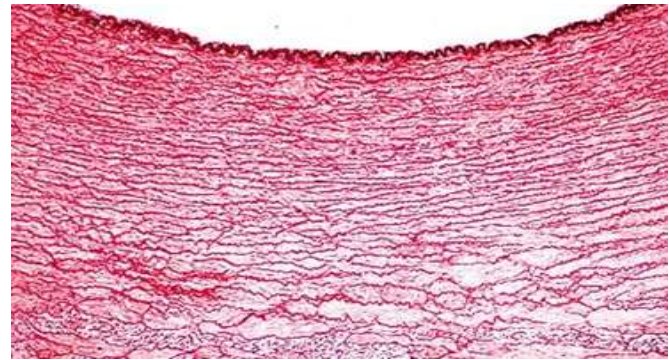
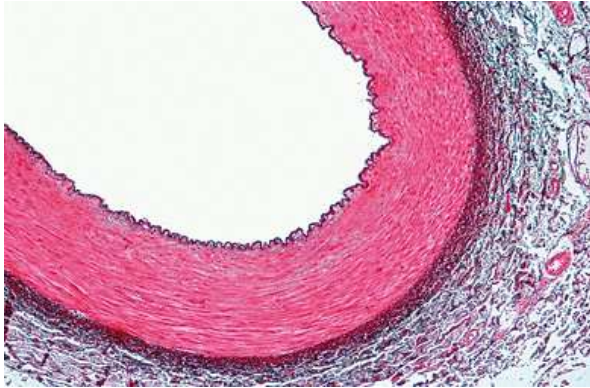
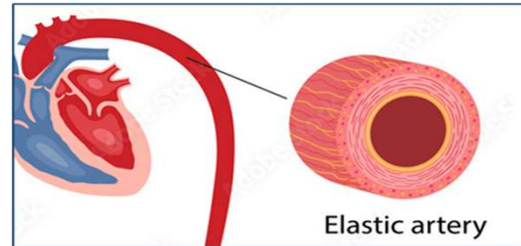
3- Small arteries (arterioles):

regulate blood flow to *capillaries*



I- Large Elastic Arteries: Aorta

large number of **elastic fibres** in **media**. **Elastic fibres > muscle**
Have thick wall & wide Lumen



1- Tunica Intima: 10%

a. Endothelium:

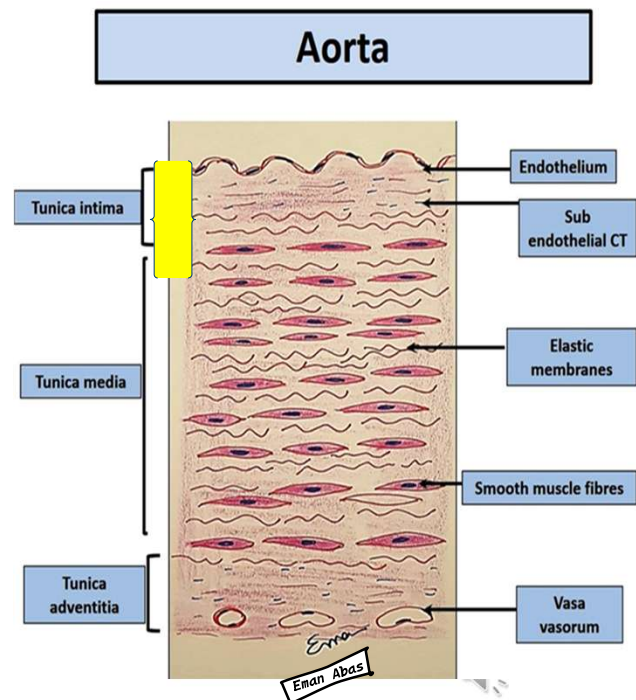
simple squamous. epith

b. Subendothelium CT:

rich in elastic fibers.

c. Internal elastic lamina:

not-clear???????



2- Tunica Media:

The thickest layer

made of:

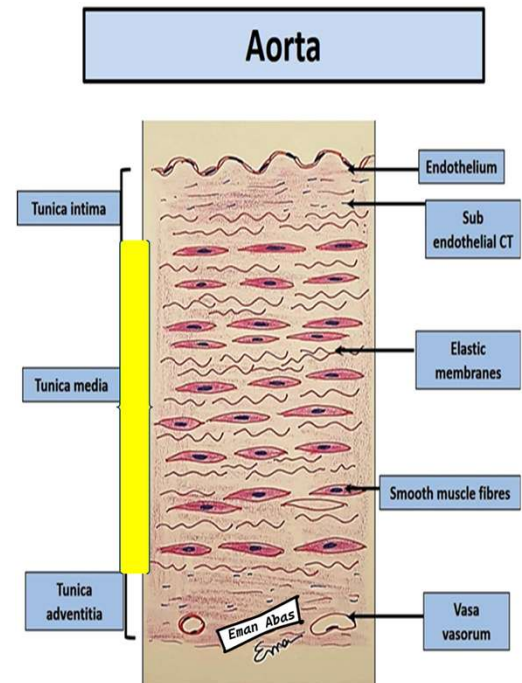
70 %

- **Elastic membranes** (circular arranged) & fenestrated = 70

+ **SMF** (circular)
=Smooth muscle fibre

+**EEL** = Not clear?

External elastic lamina



3-Tunica Adventitia:

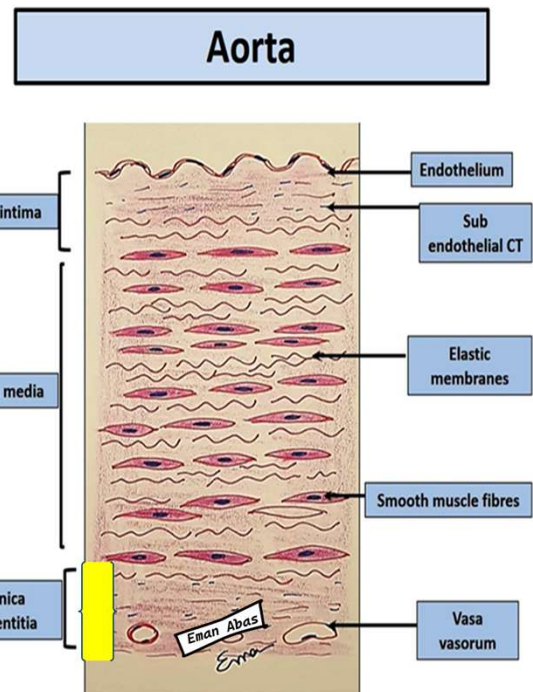
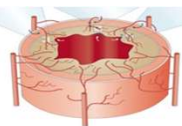
Thin coat 20 %

=loose C.T with:

- Collagen fibres .(longitudinal)& elastic fib.
- Lymphatics & nerves
- +Vasa vasorum ????????

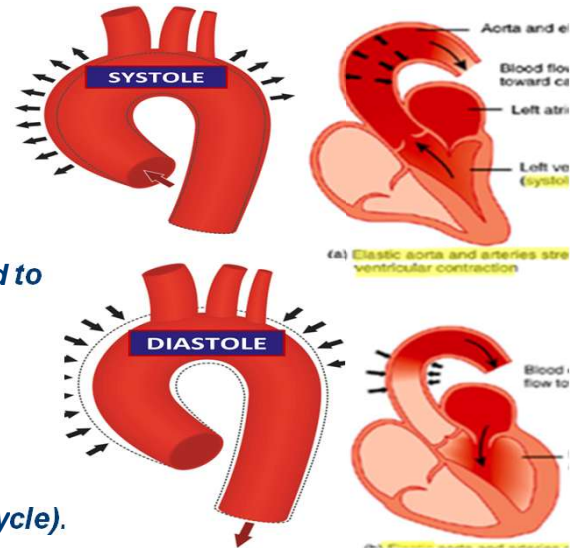
❖large arteries

to nourish the outer part of the thick vessel wall



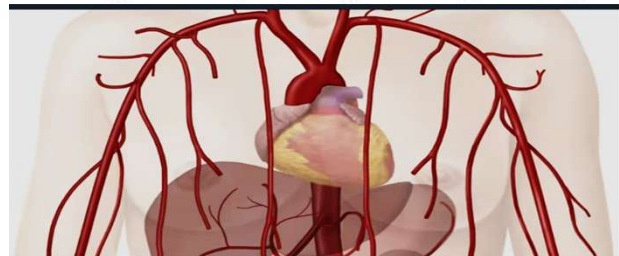
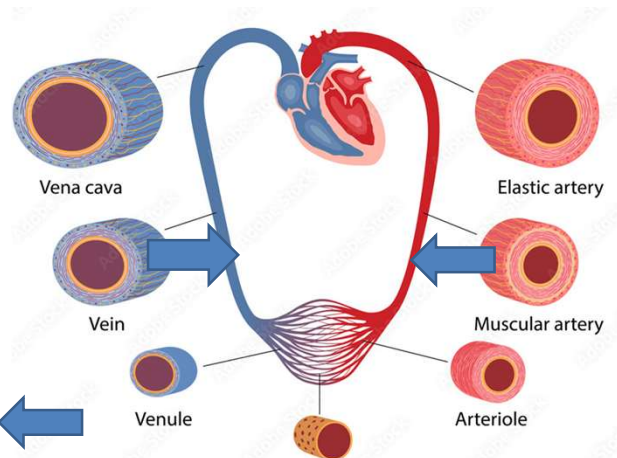
WHAT is the importance of elastic membranes in the tunica media of Aorta?

- ❖ **During systole:** *Stretch* of elastic fibres
& aorta *expand* to accommodate
large blood volume & high systolic pressure
- ❖ **During diastole:** *Recoil* of elastic fibres
return to normal size and *pushing* blood forward to
maintain continuous blood flow.
- ❖ **So, elastic fibres**
- ❖ *prevent sudden changes in blood pressure*
& blood flow during systole & diastole = (cardiac cycle).



Arteries

- Classified according to
- size & main tissue in media into:
- 1- Large elastic arteries:**
carry blood from heart.
- 2- Medium-sized muscular arteries:**
carry blood to organs.
- 3- Small arteries (arterioles):**
regulate blood flow to capillaries



A

Medium-Sized Artery & Vein

Thick
Folded
Clear IEL

Thick
Mainly SMF
+Elastic f
EEL

Thin

Thin
Not folded
No IEL

-Thin
=SMF
+V FEW
elastic f
No EEL

Thick

	Medium sized artery	Medium sized vein
Example	Radial artery	Radial vein
Function	Control blood delivery to different parts of the body	Carry blood toward heart
Wall	Thick wall.	Thin wall
Lumen	Narrow& rounded	Wide
Valves	No valves	Has valves
Tunica. Intima	Thick, folded, Clear IEL. [internal elastic lamina]	Thin, not folded, No IEL. [internal elastic lamina]
Tunica. Media	Thick, Formed mainly of many smooth muscles + some elastic fibers EEL: Has EEL. [external elastic lamina]	Thin, = smooth muscles +Very Few elastic fibers No EEL [external elastic lamina]
Tunica. Adventitia	Thin	Thick

Thank you
Prof. Eman Abas

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